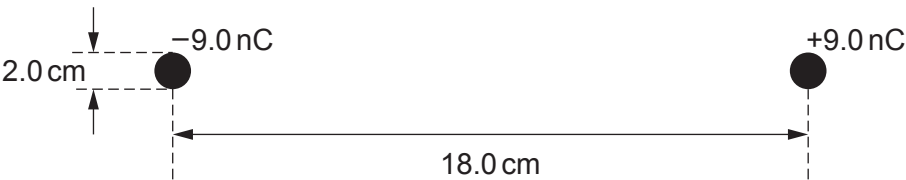
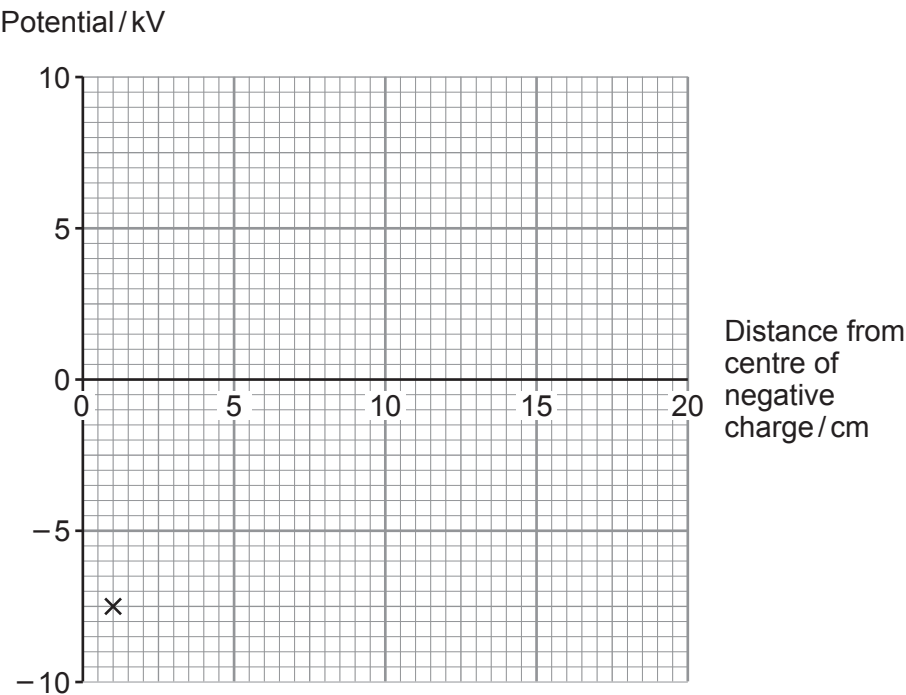


6. Two small spheres of radius 1.0 cm carry uniform charges of +9.0 nC and −9.0 nC. Their centres are separated by a distance of 18.0 cm as shown. Treat the charges as being at the centres of the spheres.



- (a) (i) Show that the potential at the surface of the negative sphere is approximately −7500 V. [3]
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- (ii) Sketch the graph for the electric potential between the two charges (the potential has been plotted for the surface of the negative charge). [3]



Examiner
only

(b) Explain why the electric field on the right-hand surface of the negative sphere is approximately $810\,000\text{ NC}^{-1}$ and directed towards the left.

[3]

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